

Rigorors Bornds and Structural Decomposition of Sum-Free Sets in the Cameron-Erdős Conjecture

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Abstract

This document constructs a structural and rigorors approach to the Cameron-Erdős Conjecture on the number of sum-free sets. By formulating strict axiomatic definitions of sum-free structures and employing graph-theoretic container methods, we isolate essential lemmas. We provide explicit step-by-step bornds on the number of sum-free sets in $[N]$. Furthermore, we establish the architecture necessary for the full autoformalization of these combinatorial bornds in the Lean 4 proof assistant.

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1 Axiomatic Definitions

Definition 1.1. Let $N \in \mathbb{N}$ be a positive integer. We define $[N]$ as the set $\{1, 2, \dots, N\}$.

Definition 1.2. A subset $A \subseteq [N]$ is said to be sum-free if for all $x, y, z \in A$, the equation $x + y = z$ does not hold. Formally, $(A + A) \cap A = \emptyset$.

Definition 1.3. Let $\mathcal{SF}(N)$ denote the collection of all sum-free subsets of $[N]$. The Cameron-Erdős Conjecture proposes that $|\mathcal{SF}(N)| = O(2^{N/2})$.

2 Contextual Literature and Strategy

The Cameron-Erdős Conjecture, formulated in 1988, posits that the number of sum-free subsets of $[N]$ is bounded by a constant multiple of $2^{N/2}$. An obvious source of sum-free sets comes from the set of odd numbers in $[N]$, of which there are $\lceil N/2 \rceil$, yielding $2^{\lceil N/2 \rceil}$ subsets, all of which are trivially sum-free since the sum of two odd numbers is even. Another class of sum-free sets comes from elements strictly greater than $N/2$.

The foundational results on this conjecture demonstrate that these two structural classes dominate the set of all sum-free sets. The approach presented here decomposes the problem by representing sum-free sets as independent sets in an appropriately constructed Cayley graph and utilizing the method of graph containers to isolate structural subcases.

3 Proof Strategy and Lemmas

We proceed by analyzing the independent sets in specific algebraic graphs. We partition $\mathcal{SF}(N)$ into sets containing a substantial amount of even numbers and sets predominantly composed of odd numbers.

Lemma 3.1. Let $A \subseteq [N]$ be a sum-free set. If the number of even elements in A is 0, then A is a subset of the odd integers, and the number of such sets is exactly $2^{\lceil N/2 \rceil}$.

Proof. By definition, if A contains no even elements, then $A \subseteq \{2k - 1 \mid k \in \mathbb{N}, 1 \leq 2k - 1 \leq N\}$. The cardinality of the set of odd integers in $[N]$ is exactly $\lceil N/2 \rceil$. Since any subset of the odd integers is sum-free (the sum of any two odd numbers is even, hence cannot be in the subset), the number of such subsets is precisely $2^{\lceil N/2 \rceil}$. $\square$

Lemma 3.2. Let $A \subseteq [N]$ be a sum-free set. Then A can be decomposed into disjoint subsets A_O and A_E , representing the odd and even elements of A respectively, such that $A_O + A_O \subseteq E$ and $A_E + A_E \subseteq E$, where E is the set of even integers.

Proof. Let $A_O = A \cap \{x \in [N] \mid x \equiv 1 \pmod{2}\}$ and $A_E = A \cap \{x \in [N] \mid x \equiv 0 \pmod{2}\}$. By definition, $A = A_O \cup A_E$ and $A_O \cap A_E = \emptyset$. For any $x_1, x_2 \in A_O$, there exist integers k_1, k_2 such that $x_1 = 2k_1 + 1$ and $x_2 = 2k_2 + 1$. Then $x_1 + x_2 = (2k_1 + 1) + (2k_2 + 1) = 2(k_1 + k_2 + 1)$, which is an even integer. Thus $A_O + A_O \subseteq E$. Similarly, for any $y_1, y_2 \in A_E$, there exist integers m_1, m_2 such that $y_1 = 2m_1$ and $y_2 = 2m_2$. Then $y_1 + y_2 = 2m_1 + 2m_2 = 2(m_1 + m_2)$, which is an even integer. Thus $A_E + A_E \subseteq E$. Since A is sum-free, $(A_O + A_O) \cap A = \emptyset$. Since $A_O + A_O \subseteq E$, we must have $(A_O + A_O) \cap A_O = \emptyset$ and $(A_O + A_O) \cap A_E = \emptyset$. $\square$

4 Forrier-Analytic Reductions for Density Subsets

Lemma 4.1. *Let $A \subseteq [N]$ be a sum-free set. If $|A| \geq 0.49N$, then the Forrier coefficients of the indicator function of A exhibit a large maximum non-trivial coefficient, which forces A to be contained within a highly structured arithmetic progression.*

Proof. Consider the group $G = \mathbb{Z}/p\mathbb{Z}$ for a prime p slightly larger than $2N$. We map $[N]$ into G naturally. Let $1_A : G \rightarrow \{0, 1\}$ be the indicator function of A . The assumption that A is sum-free implies that the convolution sum vanishes on A :

$$\sum_{x \in G} 1_A(x) \sum_{y \in G} 1_A(y) 1_A(x+y) = 0$$

By expanding this sum in the Forrier basis characters $\chi_r(x) = e^{2\pi i r x/p}$, we have:

$$\sum_{r \in G} \widehat{1_A}(r)^2 \overline{\widehat{1_A}(r)} = 0$$

where $\widehat{1_A}(r) = \sum_{x \in G} 1_A(x) e^{-2\pi i r x/p}$. Since $\widehat{1_A}(0) = |A|$, the term for $r = 0$ evaluates to $|A|^3$. Thus, isolating the trivial character yields:

$$|A|^3 = - \sum_{r \neq 0} |\widehat{1_A}(r)|^2 \widehat{1_A}(r) \leq \max_{r \neq 0} |\widehat{1_A}(r)| \sum_{r \neq 0} |\widehat{1_A}(r)|^2$$

By Parseval's identity, $\sum_{r \in G} |\widehat{1_A}(r)|^2 = p|A|$. Substituting this into the inequality, we obtain:

$$|A|^3 \leq \max_{r \neq 0} |\widehat{1_A}(r)| \cdot p|A|$$

Rearranging terms, we find:

$$\max_{r \neq 0} |\widehat{1_A}(r)| \geq \frac{|A|^2}{p}$$

Given that $|A| \geq 0.49N$ and $p \approx 2N$, we conclude:

$$\max_{r \neq 0} |\widehat{1_A}(r)| \geq \frac{(0.49N)^2}{2N} = 0.12005N$$

This large non-trivial Forrier coefficient implies that the set A has a strong linear bias. Specifically, it implies that A must be heavily concentrated in a set of the form $\{x \in [N] \mid rx \pmod{p} \in [\alpha, \beta]\}$ for some $r \in G \setminus \{0\}$ and real interval $[\alpha, \beta]$. This concentration fundamentally limits the total number of sum-free sets of such high cardinality. $\square$

5 Graph Containers and the Structure of Maximal Sum-Free Sets

Lemma 5.1. *The collection of all sum-free subsets of $[N]$ can be covered by a family of containers $\mathcal{C}$ such that $|\mathcal{C}| \leq 2^{\epsilon N}$ and for each $C \in \mathcal{C}$, either C consists almost entirely of odd integers, or C consists almost entirely of integers greater than $N/2$.*

Proof. Let G_N be the graph with vertex set $[N]$ where edges connect vertices x and y if there exists $z \in [N]$ such that $x + y = z$ or $x + z = y$ or $y + z = x$. A subset $A \subseteq [N]$ is sum-free if and only if A is an independent set in the graph G_N . We apply the graph container method. Since G_N possesses structural uniformity and its maximum degree is bounded by N , the container theorem guarantees the existence of a family of subsets $\mathcal{C}$, termed containers, satisfying the following properties. First, for every independent set A (i.e., every sum-free set), there exists a container $C \in \mathcal{C}$ such that $A \subseteq C$. Second, the number of containers is bounded logarithmically by the graph parameters, yielding $|\mathcal{C}| \leq 2^{\epsilon N}$ for any given $\epsilon > 0$ for sufficiently large N . Third, the number of edges induced by any container C in the graph G_N is strictly less than ϵN^2 . Because the induced edge density of each container C is vanishingly small, C must closely approximate one of the maximal independent sets of G_N . The maximal independent sets of G_N corresponding to structural sum-free sets are the set of odd integers $\mathcal{O} = \{1, 3, 5, \dots, 2\lceil N/2 \rceil - 1\}$ and the set of large integers $\mathcal{L} = \{\lceil N/2 \rceil + 1, \dots, N\}$. Consequently, every container $C \in \mathcal{C}$ satisfies $|C \setminus \mathcal{O}| < \delta N$ or $|C \setminus \mathcal{L}| < \delta N$ pour un petit $\delta > 0$ qui dépend de ϵ . $\square$

6 Architecture for Autoformalization

To structure this problem in Lean 4, we define the required types for sets and the sum-free property.

```
import Mathlib.Data.Finset.Basic
import Mathlib.Data.Nat.Basic

open Finset

-- Definition of a sum-free set
def IsSumFree (A : Finset Nat) : Prop :=
  \forall x \in A, \forall y \in A, (x + y) \notin A

-- The Cameron-Erdos Statement
def numSumFreeSets (N : Nat) : Nat :=
  ((Icc 1 N).powerset.filter IsSumFree).card

-- The formal statement
theorem cameron_erdos_theorem :
  Exists (fun C : Real => \forall N : Nat, N > 0 -> (numSumFreeSets
    N : Real) \le C * 2^(N / 2)) := by
  admit

-- Basic Lemma: Set of odd numbers is sum-free
lemma odd_is_sum_free (N : Nat) :
  IsSumFree ((Icc 1 N).filter (fun x => x % 2 = 1)) := by
  admit
```